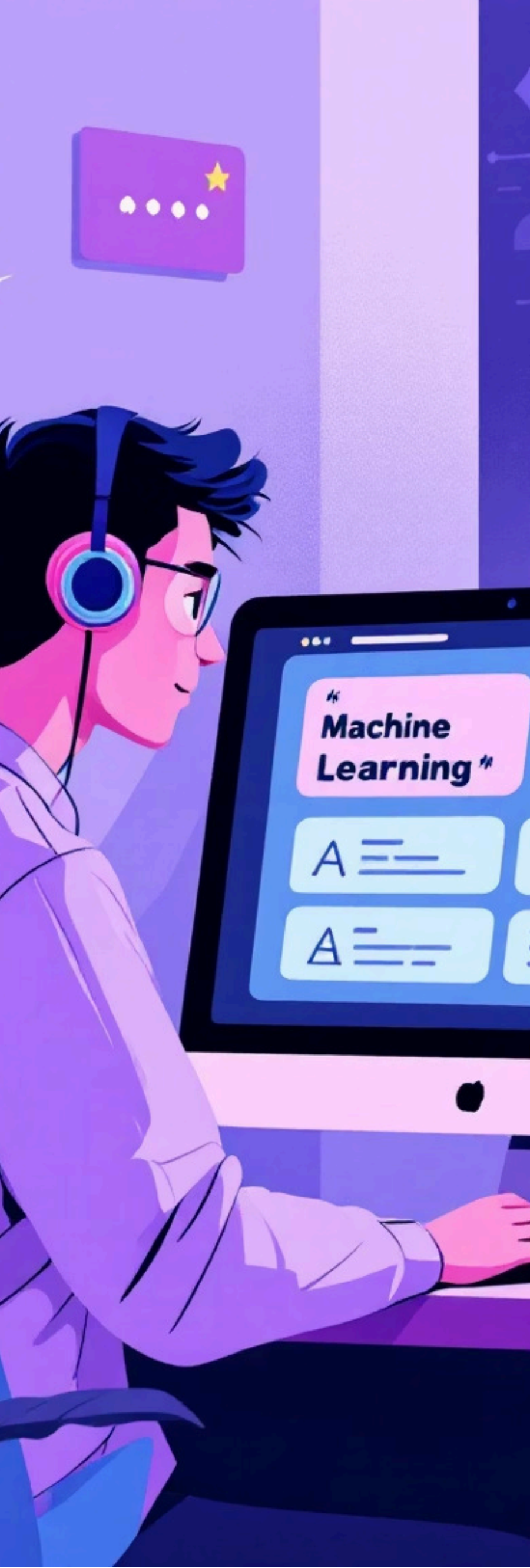




Step 5: Model Selection & Training

Model selection and training form the core of the ML lifecycle. This is where you choose the right algorithm based on your problem type and train it to learn patterns from your prepared data.

Algorithm Categories

Supervised Learning

Examples: Linear Regression, Decision Trees, Random Forest

Purpose: Prediction and classification with labelled data

Unsupervised Learning

Examples: K-Means, PCA, Hierarchical Clustering

Purpose: Pattern discovery and dimensionality reduction

Reinforcement Learning

Examples: Q-Learning, DQN, Policy Gradient

Purpose: Goal-oriented decision-making through rewards

1

Split Data

Divide into training and testing sets

2

Train Model

Feed training data to algorithm

3

Learn Patterns

Model adjusts internal parameters

4

Test Performance

Validate on unseen data

Step 6: Model Evaluation

Model evaluation is critical to ensure your model generalises well to new, unseen data. Using the right metrics helps you understand where your model excels and where it needs improvement.



Regression Tasks

- **MSE:** Mean Squared Error
- **RMSE:** Root Mean Squared Error
- **R² Score:** Coefficient of determination



Classification Tasks

- **Accuracy:** Overall correctness
- **Precision:** Positive prediction accuracy
- **Recall:** True positive detection rate
- **F1-Score:** Harmonic mean of precision and recall



Clustering Tasks

- **Silhouette Score:** Cluster separation quality
- **Davies-Bouldin Index:** Cluster compactness



Key Goal: Ensure your model generalises well and avoids overfitting—where it performs brilliantly on training data but poorly on real-world data.

Step 7: Hyperparameter Tuning



Hyperparameter tuning is the process of optimising the configuration settings of your ML algorithm to achieve the best possible performance. Unlike model parameters learned during training, hyperparameters are set before the learning process begins.

Popular Tuning Techniques



Grid Search

Exhaustively searches through a manually specified subset of hyperparameter space. Systematic but computationally expensive for large parameter grids.



Random Search

Randomly samples hyperparameter combinations. More efficient than grid search and often finds good solutions faster.



Bayesian Optimisation

Uses probabilistic models to intelligently select the next hyperparameters to try. Most efficient for expensive model training.

Impact: Fine-tuning reduces bias and variance issues, leading to improved model accuracy and better generalisation on unseen data.

Step 8: Model Deployment

Deployment bridges the gap between your trained model and real-world applications. This step transforms your ML model from an experimental notebook into a production-ready system that delivers value to end users.

1

Save Model

Serialise using .pkl, .joblib, or ONNX format

2

Create API

Build REST endpoints with Flask or FastAPI

3

Deploy to Cloud

Use AWS SageMaker, Azure ML, or GCP AI Platform

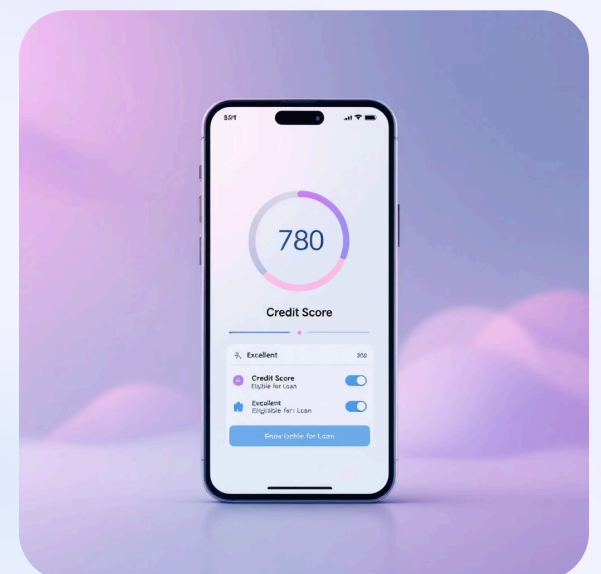
4

Integrate

Connect with production applications

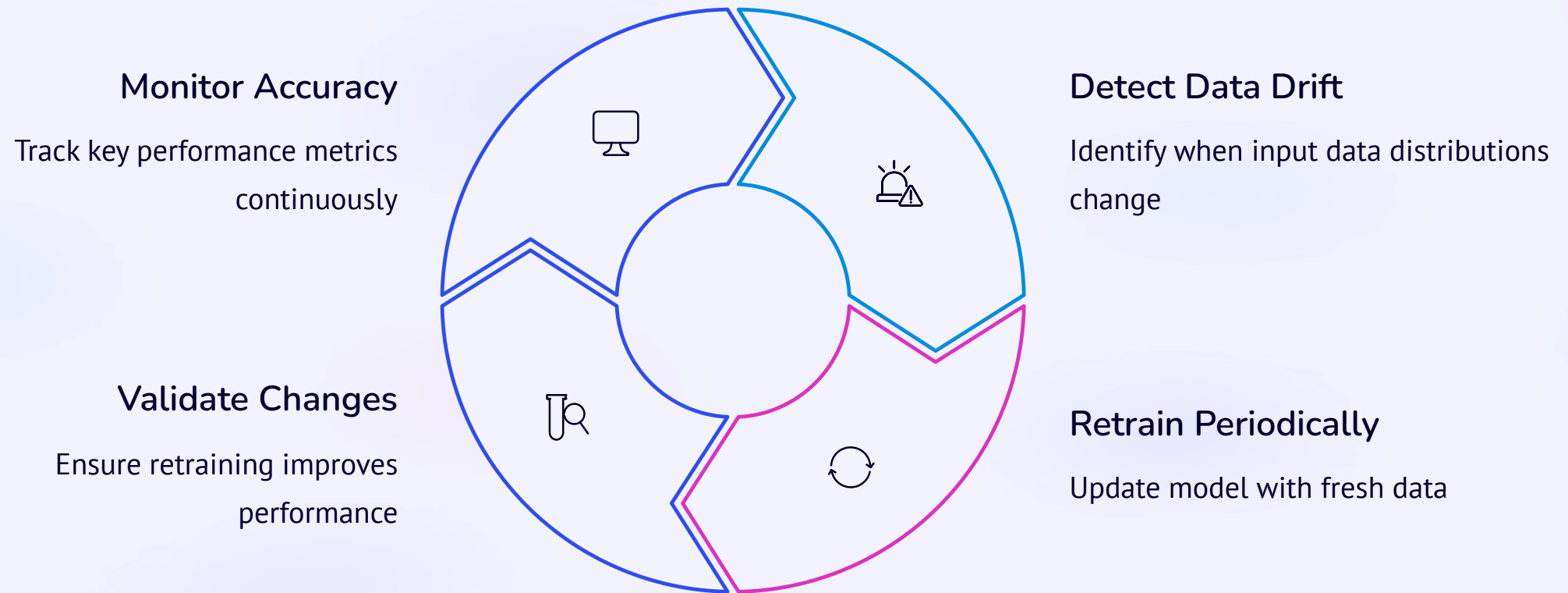
Real-World Example

A credit eligibility prediction model integrated into a banking application. The model receives customer data via API, processes it in real-time, and returns creditworthiness scores that inform loan approval decisions instantly.



Step 9: Monitoring & Maintenance

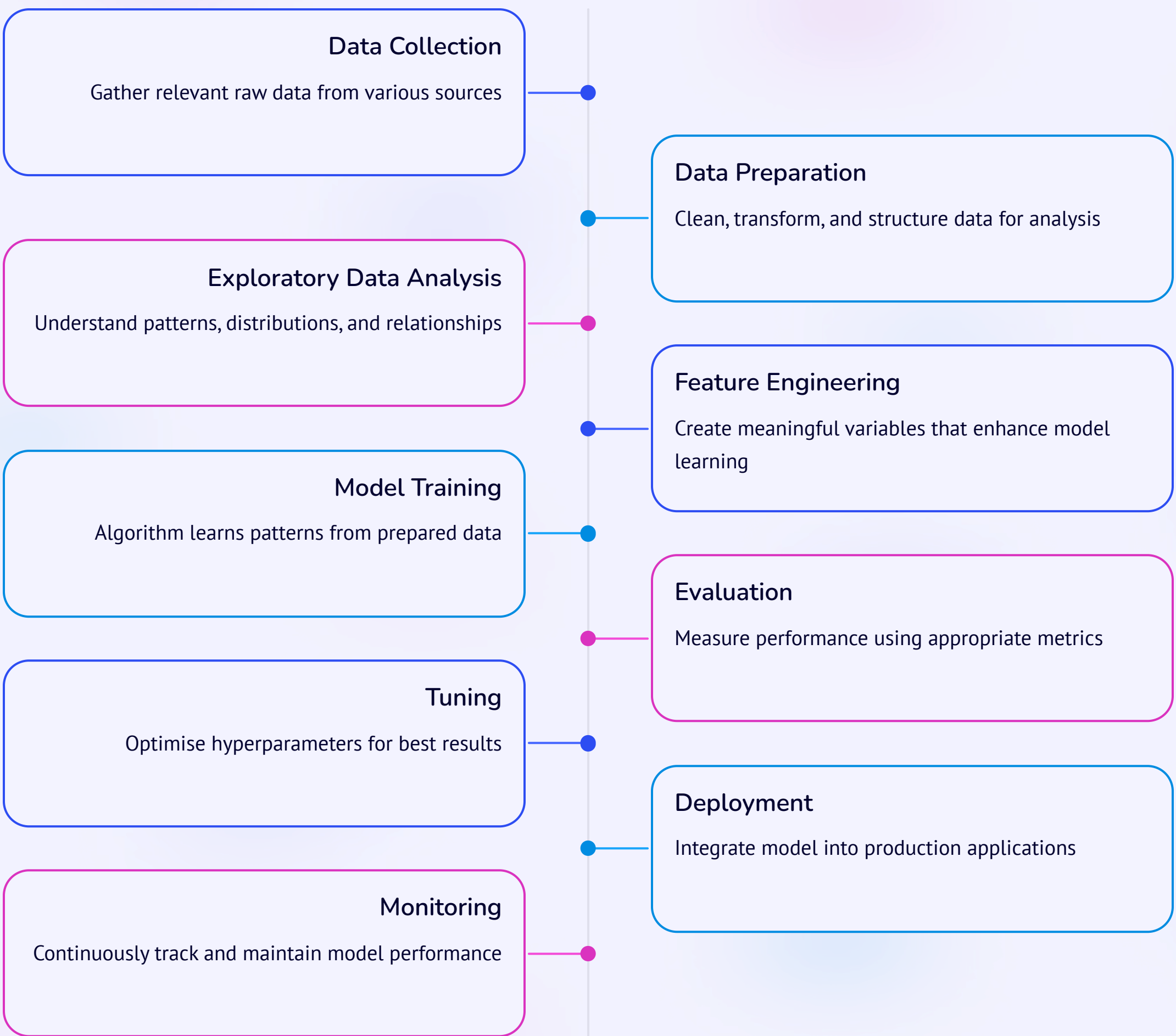
Deployment isn't the end—it's the beginning of an ongoing process. Real-world data evolves, and your model's performance must be continuously monitored to ensure it remains accurate and relevant over time.



❏ **Why This Matters:** Data patterns shift due to seasonality, market changes, or user behaviour evolution. Without monitoring, your once-accurate model can quietly degrade, leading to poor decisions and lost business value.

The Complete ML Lifecycle

The machine learning lifecycle is a systematic framework that guides you from raw data to production-ready models. Understanding each stage ensures reliable, maintainable, and effective ML systems.



Key Roles in the ML Process

Building successful ML systems requires collaboration across multiple roles. Each brings specialised expertise that contributes to different stages of the lifecycle.



Data Scientist

Leads data preprocessing, exploratory analysis, feature engineering, and model development. Bridges business problems with statistical solutions and interprets model outputs.



ML Engineer

Focuses on model deployment, scalability, and optimisation. Builds production pipelines, manages infrastructure, and ensures models run efficiently at scale.



Data Engineer

Designs and maintains data pipelines and storage solutions. Ensures data quality, availability, and accessibility for downstream ML processes.



Business Analyst

Defines problems from business perspective, interprets model results for stakeholders, and ensures ML solutions align with organisational goals and deliver measurable value.

Quick Recap: ML Fundamentals

What is Machine Learning?

A system that learns patterns from data without being explicitly programmed. It identifies relationships and makes predictions based on historical examples.

Why Use Machine Learning?

To automate complex decision-making, predict future outcomes, discover hidden patterns, and scale solutions that would be impossible to code manually.

The ML Lifecycle

A systematic journey from data collection through model training, evaluation, and deployment, followed by continuous monitoring and maintenance.

"Machine Learning is not magic—it's data, logic, and iteration at scale."

Success in ML comes from disciplined execution of each lifecycle stage, continuous learning from failures, and maintaining a data-driven mindset throughout the process.



Key Takeaways

○ Data-Driven Automation

ML enables organisations to automate complex decisions and generate accurate predictions at scale, transforming raw data into actionable business intelligence.

○ Systematic Approach

Following a clear lifecycle—from data collection through deployment—ensures systematic, reliable, and reproducible model building. Each stage builds upon the previous one.

○ Continuous Improvement

Ongoing monitoring keeps models relevant as data evolves. Regular retraining and performance tracking prevent model degradation and maintain accuracy over time.

📌 **Next Steps:** Apply these concepts to your next ML project. Start with a well-defined problem, follow the lifecycle systematically, and remember that iteration is key to success.